

Implementation of Puppet Modules

Problem Scenario

An organization is managing multiple Linux servers using Puppet. The application configuration for `myapp` is currently written in a standalone Puppet class.

As the infrastructure grows, the administrator wants to package the application configuration into a **Puppet Module** so that the same configuration can be reused across multiple servers and environments.

The module should manage:

- `/home/Man` directory
- `/home/Man/myapp` directory
- `/home/Man/myapp/config.txt`
- Configuration file content

Requirements

Hardware Requirements

- One Puppet Master machine
- One Puppet Agent machine
- Minimum 2 GB RAM recommended per machine
- Network connectivity between Master and Agent

Software Requirements

- Ubuntu Linux
- Puppet Server
- Puppet Agent
- OpenSSH
- Terminal/Shell

Pre-Requirements

Before starting this practical, ensure that:

5.1 Puppet Master is running

On Master:

```
sudo systemctl status puppetserver
```

Expected:

Active: active (running)

5.2 Puppet Agent is running

On Agent:

```
sudo systemctl status puppet
```

Expected:

Active: active (running)

5.3 Master-Agent communication is working

On Agent:

```
sudo /opt/puppetlabs/puppet/bin/puppet agent --test
```

Expected:

Notice: Requesting catalog from puppet-master:8140

Notice: Catalog compiled by puppet-master

Notice: Applied catalog in ...

5.4 Production environment exists

On Master:

```
sudo ls /etc/puppetlabs/code/environments/production
```

Expected:

manifests

5.5 Previous Class Practical

The previous `myapp` class should already be working:

production/

└─ manifests/

 └─ site.pp

 └─ myapp.pp

And `site.pp` should contain:

include myapp

On master:

```
ubuntu@puppet-master:~$ sudo cat /etc/puppetlabs/code/environments/production/manifests/myapp.pp
class myapp {

  file { ['/home/Man':
    ensure => directory,
  ]

  file { ['/home/Man/myapp':
    ensure => directory,
    require => File['/home/Man'],
  ]

  file { ['/home/Man/myapp/config.txt':
    ensure => file,
    content => "Application managed by Puppet\n",
    require => File['/home/Man/myapp'],
  ]

}
ubuntu@puppet-master:~$
```

```
ubuntu@puppet-master:~$ sudo mkdir -p /etc/puppetlabs/code/environments/production/modules/myapp/manifests
ubuntu@puppet-master:~$ sudo ls -R /etc/puppetlabs/code/environments/production/modules/myapp
/etc/puppetlabs/code/environments/production/modules/myapp:
manifests

/etc/puppetlabs/code/environments/production/modules/myapp/manifests:
ubuntu@puppet-master:~$
```

```
ubuntu@puppet-master:~$ sudo nano /etc/puppetlabs/code/environments/production/modules/myapp/manifests/init.pp
ubuntu@puppet-master:~$ sudo cat /etc/puppetlabs/code/environments/production/modules/myapp/manifests/init.pp
class myapp {

  file { ['/home/Man':
    ensure => directory,
  ]

  file { ['/home/Man/myapp':
    ensure => directory,
    require => File['/home/Man'],
  ]

  file { ['/home/Man/myapp/config.txt':
    ensure => file,
    content => "Application managed by Puppet\n",
    require => File['/home/Man/myapp'],
  ]

}
ubuntu@puppet-master:~$
```

On Agent:

```
Notice: Applied catalog in 0.03 seconds
ubuntu@puppet-agent:~$ sudo /opt/puppetlabs/puppet/bin/puppet agent --test
Info: Refreshing CA certificate
Info: CA certificate is unmodified, using existing CA certificate
Info: Refreshing CRL
Info: CRL is unmodified, using existing CRL
Info: Using environment 'production'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Warning: Fact value '-----BEGIN CERTIFICATE-----
MIIGATCCBQmGA1Bg1QCOpGr1P+KwbzQ+MEWm+DQANBgkqhkiG9w0BAQsFADA8
MQAwCQYDVQQGEwJVUzEPMA0GA1UEChMGMQWlhem9uMRwwGyYDVQQDEsNBhWF6b24g
U1NBID1wNDggTTA0MB4XDTI2MDgwMzAwMDAwMFoXDTI3MDIwNjEzNTk1OVowNjE0
MDIGAlURAAxMrbWfuYkd1ZC1zc2gtc2lnbmVybW9uYXNjaW50b3R0b3R0b3R0b3R0
-----END CERTIFICATE-----' with the value length: '5363' exceeds the value length limit: 4096
Notice: Requesting catalog from puppet-master:8140 (172.31.3.190)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-agent.ap-south-1.compute.internal
Info: Applying configuration version '1789837999'
Notice: Applied catalog in 0.02 seconds
ubuntu@puppet-agent:~$
```

```
ubuntu@puppet-agent:~$ ls -l /home/Man
total 4
drwxr-xr-x 2 root root 4096 Sep 19 16:39 myapp
ubuntu@puppet-agent:~$ ls -l /home/Man/myapp
total 4
-rw-r--r-- 1 root root 30 Sep 19 16:39 config.txt
ubuntu@puppet-agent:~$ cat /home/Man/myapp/config.txt
Application managed by Puppet
ubuntu@puppet-agent:~$
```

Final Directory Structure

Puppet Master

/etc/puppetlabs/code/environments/production/

```
|
|—— manifests/
|   |—— site.pp
|
|—— modules/
|   |—— myapp/
|       |—— manifests/
|           |—— init.pp
```